

Series 1: Basic elements

Exercise 1: Progressive Discovery of Algorithmics with Blockly Maze (to be done in the Lab Session)

Algorithmics allows us to describe, in the form of precise instructions, the steps necessary to solve a problem.

In this exercise, you will use the **Blockly Maze** environment (<https://blockly.games/maze>) to progressively discover the fundamental concepts of algorithmics.

Tasks:

1. Sequence of instructions

- Play the first levels of the game (straight-line movement).
- Question: Which instructions are repeated in your solution?
- Question: How can you translate these instructions into pseudo-code?

2. Repetition (loops)

- At a certain level, it becomes tedious to repeat the *Move forward* instruction many times.
- Question: Which algorithmic structure helps to avoid this repetition?
- Question: Reformulate your solution in pseudo-code using a loop.

3. Condition (if...then...else)

- In later levels, the character encounters an obstacle.
- Question: What decision must the character take to continue moving?
- Question: How can you translate this logic into pseudo-code (conditional structure)?

4. Loops with conditions (while / repeat until)

- At a certain stage, the character must keep moving until reaching the flag, but the exact number of steps is unknown.
- Question: Which algorithmic structure corresponds to this type of situation?
- Question: Translate the solution into pseudo-code using **While** or **Repeat until**.

5. Reflective analysis

- Compare your solutions with and without loops/conditions.
- Question: Why is an optimized solution more interesting than a naïve one?
- Question: What similarities do you notice between solving problems in Blockly Maze and the formal algorithms you will study in this course?

Exercise 2: Measurement problems with cans consist of using cans of different capacities to obtain specific quantities. Finding a solution to the following distribution problems:

1. **With 3 cans:** We have **one can** of "A" liters of juice and **two other empty cans**, one of "B" liters and one of "C" liters. How can we divide the juice into two equal parts of " $A/2$ " liters using these cans?

No. Example	1	2	3	4	5
A	20	18	30	32	40
B	13	11	19	21	25
C	7	7	11	11	15

2. **With 4 cans:** We have **one can** of "A" liters of juice and **three other empty cans**, one of "B" liters, one of "C" liters and one of "D" liters. How can we divide the juice into two equal parts of " $A/2$ " liters using these cans?

No. Example	1	2	3	4
A	28	36	40	36
B	15	24	25	20
C	9	15	20	10
D	3	9	5	5

Exercise 3: The principle of **river crossing problems** is to transport individuals or objects from one point to another (by crossing a river), while respecting specific constraints that make resolution non-trivial. Constraints vary: some individuals cannot be left together, the number of people transported at a time is limited, or some people cannot navigate or move alone. Finding a solution to the following crossing issues:

1. A father, mother and two children must cross a river using a boat, but the boat can only carry two people at a time. Only father and mother know how to row.
2. Three thieves and three guards must cross a river using a boat. The boat can only carry two people at a time. If the thieves are left alone with an outnumbered guard, they will attack him.
3. Three explorers and three cannibals must cross a river using a boat, which can only carry two people at a time. If at any time cannibals are in the majority compared to explorers on a shore (whether arriving or departing), they will eat the explorers.
4. A farmer has to cross a river with a fox, a chicken and a bag of corn. However, the boat can only hold the farmer and only one of the three items. If the fox is left alone with the chicken, it will eat it, and if the chicken is left alone with the bag of corn, it will eat the corn.
5. The farmer has to cross the river with a wolf, a chicken, a bag of corn and a goat. The main constraints are:
 - a. The wolf should not be left alone with the chicken.

- b. The chicken should not be left alone with the bag of corn.
- c. The goat should not be left alone with the corn, as it will eat it.

Exercise 04: For each problem, specify the actions to be taken to solve it (in the form of analysis) as well as the input data and the results obtained/returned.

- 1) Calculate the speed (in km/h) from a duration (in hours [h]) and a distance (in kilometers [km])
- 2) Calculate the perimeter (in centimetres [cm]) and area (in cm^2) of a circle determined by its radius
- 3) Calculate the distance between two points x, y of respective coordinates (x1, x2), and (y1, y2).
- 4) Given a time in hours h, minute m, and seconds s, transform this time into seconds.
- 5) Given a time in seconds, transform this time into h, m, s.
- 6) Convert a temperature T_c , expressed in degrees Celsius, to Fahrenheit degrees T_f , knowing that: $T_f = T_c * 1.8 + 32$.
- 7) Let "Md" be an amount in Algerian Dinar and "tc" the exchange rate in Euro, calculate the equivalent amount in Euro.
- 8) Perform the permutation between two integer variables A and B
- 9) Give the maximum of two integers,
- 10) Check if an integer "a" divides an integer "b",
- 11) Calculate the sum of the first n integers ($1+2+3+\dots+n$),
- 12) Calculate the factorial of a natural number n ($n! = 1*2*3*\dots*n$).
- 13) Calculate the elevation of a real number x to the power of an integer n: x^n ,
- 14) Calculate a student's average for a set of subjects.
- 15) Calculate the integer square root of a positive integer,
- 16) Determine all the divisors of an integer x.
- 17) Calculate the sum of all divisors of an integer x,
- 18) Determine the smallest divisor of an integer greater than 1,
- 19) Determine the largest divisor of an integer greater than 1,
- 20) Solving a second-degree equation of the form $Ax^2+Bx+C=0$